ADV: Functions (Adv), F1 Working with Functions (Adv)

Quadratics, Circles and Other Non-Linear (Y11)

Teacher: Ms Najem

Exam Equivalent Time: 67.5 minutes *(based on HSC allocation of 1.5 minutes approx. per mark)*

Questions

1. Functions, 2ADV F1 2013 HSC 1 MC

What are the solutions of $2x^2 - 5x - 1 = 0$?

(A) $x = \dfrac{-5 \pm \sqrt{17}}{4}$

(B) $x = \dfrac{5 \pm \sqrt{17}}{4}$

(C) $x = \dfrac{-5 \pm \sqrt{33}}{4}$

(D) $x = \dfrac{5 \pm \sqrt{33}}{4}$

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2. Functions, 2ADV F1 2014 HSC 6 MC

Which expression is a factorisation of $8x^3 + 27$?

(A) $(2x - 3)(4x^2 + 12x - 9)$

(B) $(2x + 3)(4x^2 - 12x + 9)$

(C) $(2x - 3)(4x^2 + 6x - 9)$

(D) $(2x + 3)(4x^2 - 6x + 9)$

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3. Functions, 2ADV F1 2017 HSC 2 MC

Which expression is equal to $3x^2 - x - 2$?

(A) $(3x - 1)(x + 2)$

(B) $(3x + 1)(x - 2)$

(C) $(3x - 2)(x + 1)$

(D) $(3x + 2)(x - 1)$

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4. Functions, 2ADV F1 2013 HSC 3 MC

Which inequality defines the domain of the function $f(x) = \dfrac{1}{\sqrt{x+3}}$?

(A) $x > -3$

(B) $x \geq -3$

(C) $x < -3$

(D) $x \leq -3$

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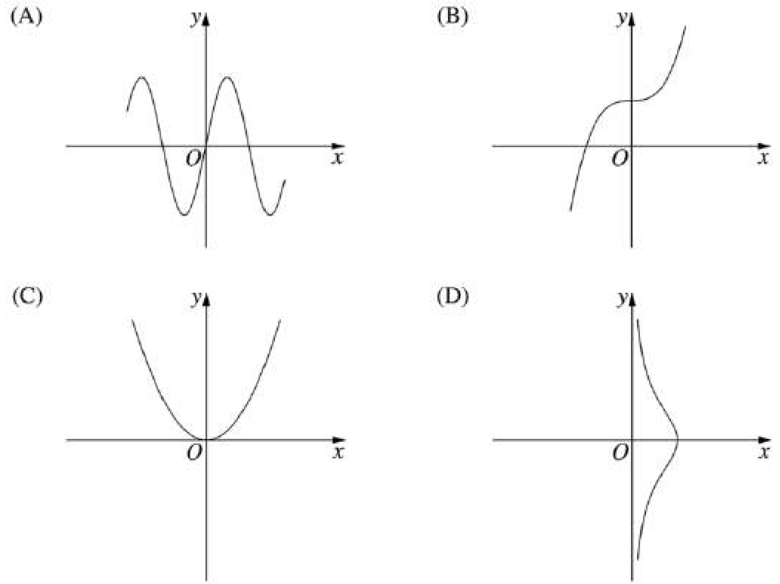
5. Functions, 2ADV F1 2016 HSC 4 MC

Which diagram shows the graph of an odd function?



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6. Functions, 2ADV F1 2006 HSC 1b

Factorise $2x^2 + 5x - 3$. *(2 marks)*

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7. Functions, 2ADV F1 2007 HSC 1e

Factorise $2x^2 + 5x - 12$. *(2 marks)*

8. Functions, 2ADV F1 2016 HSC 11e

Find the points of intersection of $y = -5 - 4x$ and $y = 3 - 2x - x^2$. *(3 marks)*

9. Functions, 2ADV F1 2010 HSC 1a

Solve $x^2 = 4x$. *(2 marks)*

10. Functions, 2ADV F1 2012 HSC 11a

Factorise $2x^2 - 7x + 3$ *(2 marks)*

11. Functions, 2ADV F1 2014 HSC 11b

Factorise $3x^2 + x - 2$. *(2 marks)*

12. Functions, 2ADV F1 2015 HSC 11b

Factorise fully $3x^2 - 27$. *(2 marks)*

13. Functions, 2ADV F1 SM-Bank 33

i. State the domain and range of $y = -\sqrt{12 - x^2}$. *(2 marks)*

ii. Sketch the graph. *(1 mark)*

14. Functions, 2ADV F1 2016 HSC 11a

Sketch the graph of $(x-3)^2 + (y+2)^2 = 4$. *(2 marks)*

15. Functions, 2ADV F1 SM-Bank 22

Worker A picks a bucket of blueberries in a hours. Worker B picks a bucket of blueberries in b hours.

i. Write an algebraic expression for the fraction of a bucket of blueberries that could be picked in one hour if A and B worked together. *(2 marks)*

ii. What does the reciprocal of this fraction represent? *(1 mark)*

16. Functions, 2ADV F1 2008 HSC 1c

Simplify $\dfrac{2}{n} - \dfrac{1}{n+1}$. *(2 marks)*

17. Functions, 2ADV F1 SM-Bank 21

Simplify $\left(\dfrac{p}{q}\right)^3 \div \left(pq^{-2}\right)$. *(2 marks)*

18. Functions, 2ADV F1 SM-Bank 23

Find the values of k for which the expression $x^2 - 3x + (4 - 2k)$ is always positive. *(3 marks)*

19. Functions, 2ADV F1 SM-Bank 26

Fuifui finds that for a certain species of eel, the mass of the eel is directly proportional to the cube of its length.

An eel of this species has a length of 25 cm and a mass of 4350 grams.

What is the expected length of an eel with a mass of 6.2 kg? Give your answer to one decimal place. *(3 marks)*

20. Functions, 2ADV F1 SM-Bank 27

The stopping distance of a car on a certain road, once the brakes are applied, is directly proportional to the square of the speed of the car when the brakes are first applied.

A car travelling at 70 km/h takes 58.8 metres to stop.

How far does it take to stop if it is travelling at 105 km/h? *(3 marks)*

21. Functions, 2ADV F1 SM-Bank 32

Find the centre and radius of the circle with the equation

$x^2 - 12x + y^2 + 2y - 12 = 0$ *(2 marks)*

22. Functions, 2ADV F1 2010 HSC 1c

Write down the equation of the circle with centre (–1, 2) and radius 5. *(1 mark)*

23. Functions, 2ADV F1 2010 HSC 1g

Let $f(x) = \sqrt{x-8}$. What is the domain of $f(x)$? *(1 mark)*

24. Functions, 2ADV F1 2017 HSC 11h

Find the domain of the function $f(x) = \sqrt{3-x}$. *(2 marks)*

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Worked Solutions

1. Functions, 2ADV F1 2013 HSC 1 MC

$$2x^2-5x-1=0$$

$$\text{Using } x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$\begin{aligned} x &= \frac{5\pm\sqrt{(-5)^2-4\times2\times(-1)}}{2\times2} \\ &= \frac{5\pm\sqrt{25+8}}{4} \\ &= \frac{5\pm\sqrt{33}}{4} \end{aligned}$$

$\Rightarrow D$

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2. Functions, 2ADV F1 2014 HSC 6 MC

$$\begin{aligned} &8x^3+27 \\ &= (2x)^3+3^3 \\ &= (2x+3)\left(4x^2-6x+9\right) \end{aligned}$$

$\Rightarrow D$

COMMENT: Factorising a cubic has been notably left off the new 2020 exam reference sheet, although we still recommend knowing it!

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3. Functions, 2ADV F1 2017 HSC 2 MC

$$\begin{aligned} &3x^2-x-2 \\ &= (3x+2)(x-1) \end{aligned}$$

$\Rightarrow D$

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4. Functions, 2ADV F1 2013 HSC 3 MC

$$\text{Given } f(x)=\frac{1}{\sqrt{x+3}}$$

$$\begin{aligned} \text{We know } (x+3) &> 0 \\ x &> -3 \end{aligned}$$

$\therefore$ The domain of $f(x)$ is $f(x)>-3$

$\Rightarrow A$

Worked Solutions

5. Functions, 2ADV F1 2016 HSC 4 MC

♦ Mean mark 38%.

Odd functions occur when:

$f(x) = -f(x)$

Graphically, this occurs when a function has symmetry when rotated 180° about the origin.

$\Rightarrow A$

6. Functions, 2ADV F1 2006 HSC 1b

$$\begin{aligned}&2x^2+5x-3\\&\quad=(2x-1)(x+3)\end{aligned}$$

7. Functions, 2ADV F1 2007 HSC 1e

$$\begin{aligned}&2x^2+5x-12\\&\quad=(2x-3)(x+4)\end{aligned}$$

8. Functions, 2ADV F1 2016 HSC 11e

$y = 3-2x-x^2$

Substitute $y=-5-4x$ into equation

$$\begin{aligned}-5-4x&=3-2x-x^2\\x^2-2x-8&=0\\(x-4)(x+2)&=0\end{aligned}$$

$\therefore x = 4 \text{ or } -2$

When $x=4,\ \ y=-5-4(4)=-21$

When $x=-2,\ \ y=-5-4(-2)=3$

$\therefore$ Intersection at $(4,-21)$ and $(-2,3)$

9. Functions, 2ADV F1 2010 HSC 1a

$$\begin{aligned}x^2&=4x\\x^2-4x&=0\\x(x-4)&=0\end{aligned}$$

$\therefore\ x = 0 \text{ or } 4$

10. Functions, 2ADV F1 2012 HSC 11a

$$\begin{aligned}&2x^2-7x+3\\&\quad=(2x-1)(x-3)\end{aligned}$$

STRATEGY: Check your answer by expanding factors.

11. Functions, 2ADV F1 2014 HSC 11b

$$\begin{aligned}&3x^2+x-2\\&\quad=(3x-2)(x+1)\end{aligned}$$

12. Functions, 2ADV F1 2015 HSC 11b

$$\begin{aligned}3x^2-27&=3(x^2-9)\\&=3(x+3)(x-3)\end{aligned}$$

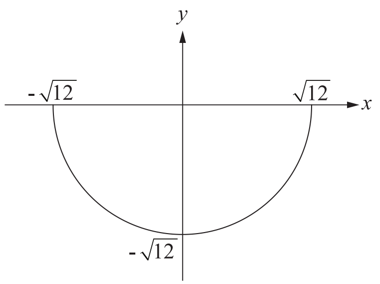
13. Functions, 2ADV F1 SM-Bank 33

i. $y = -\sqrt{12 - x^2}$

Domain: $-\sqrt{12} \le x \le \sqrt{12}$

Range: $-\sqrt{12} \le y \le 0$

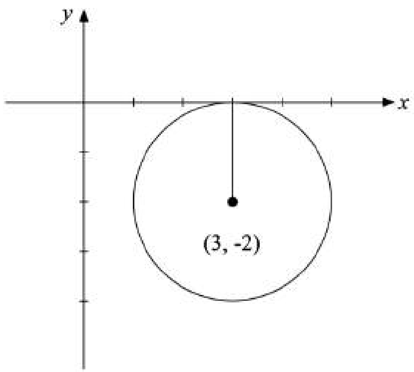
ii.



14. Functions, 2ADV F1 2016 HSC 11a

$(x-3)^2 + (y+2)^2 = 4$ is a circle,

centre $(3, -2)$, radius 2.



15. Functions, 2ADV F1 SM-Bank 22

i. In one hour:

Worker A picks $\dfrac{1}{a}$ bucket.

Worker B picks $\dfrac{1}{b}$ bucket.

COMMENT: Note that the question asks for "a fraction".

$\therefore$ Fraction picked in 1 hour working together

$$= \frac{1}{a} + \frac{1}{b}$$

$$= \frac{a+b}{ab}$$

ii. The reciprocal represents the number of hours it would take to fill one bucket, with A and B working together.

16. Functions, 2ADV F1 2008 HSC 1c

$$\frac{2}{n} - \frac{1}{n+1}$$

$$= \frac{2(n+1) - 1(n)}{n(n+1)}$$

$$= \frac{2n+2-n}{n(n+1)}$$

$$= \frac{n+2}{n(n+1)}$$

17. Functions, 2ADV F1 SM-Bank 21

$$\left(\frac{p}{q}\right)^3 \div (pq^{-2}) = \frac{p^3}{q^3} \div \frac{p}{q^2}$$

$$= \frac{p^3}{q^3} \times \frac{q^2}{p}$$

$$= \frac{p^2}{q}$$

18. Functions, 2ADV F1 SM-Bank 23

$x^2-3x+(4-2k)>0$

$x^2-3x+(4-2k)=0$ is a concave up parabola

$\Rightarrow$ Always positive (no roots) if $\Delta<0$

$$b^2-4ac<0$$

$$\begin{aligned}(-3)^2-4\cdot 1\cdot(4-2k)&<0\\ 9-16+8k&<0\\ 8k&<7\\ k&<\frac{7}{8}\end{aligned}$$

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19. Functions, 2ADV F1 SM-Bank 26

Mass $\propto$ length^3

$m=kl^3$

Find k:

$$\begin{aligned}4350&=k\times 25^3\\ k&=\frac{4350}{25^3}\\ &=0.2784\end{aligned}$$

When l when $m=6200$:

$$\begin{aligned}6200&=0.2784\times l^3\\ l^3&=\frac{6200}{0.2784}\\ \therefore l&=28.13\ldots\\ &=28.1\text{ cm (to 1 d.p.)}\end{aligned}$$

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20. Functions, 2ADV F1 SM-Bank 27

Let $d=$ stopping distance

$d\propto s^2$

$d=ks^2$

Find k,

$$\begin{aligned}58.8&=k\times 70^2\\ k&=\frac{58.8}{70^2}\\ &=0.012\end{aligned}$$

Find d when $s=105$:

$$\begin{aligned}d&=0.012\times 105^2\\ &=132.3\text{ metres}\end{aligned}$$

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21. Functions, 2ADV F1 SM-Bank 32

$$\begin{aligned}x^2-12x+y^2+2y-12&=0\\ (x-6)^2+(y+1)^2-36-1-12&=0\\ (x-6)^2+(y+1)^2&=49\end{aligned}$$

$\therefore$ Centre $(6,-1)$

$\therefore$ Radius $=7$

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22. Functions, 2ADV F1 2010 HSC 1c

Circle with centre $(-1,2)$, $r=5$

$$(x+1)^2+(y-2)^2=25$$

MARKER'S COMMENT: Expanding this equation is not necessary!

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23. Functions, 2ADV F1 2010 HSC 1g

$f(x)=\sqrt{x-8}$

Domain where

$$\begin{aligned}(x-8)&\geq 0\\ x&\geq 8\end{aligned}$$

♦ *Mean mark 49%.*
MARKER'S COMMENT: $x>8$ was a common incorrect answer.

24. Functions, 2ADV F1 2017 HSC 11h

Solution 1

Domain of $f(x) = \sqrt{3 - x}$

$$3 - x \geq 0$$
$$x \leq 3$$

Note domain can also be expressed as: $(-\infty, 3]$